

$\Sigma$

## A Library for ANSI Common Lisp

<https://github.com/cgore/sigma/>

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Sunday, August 9<sup>th</sup>, AD 2026



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# Introduction

The  $\Sigma$  library is a generic library of mostly random useful code for ANSI Common Lisp. It is currently only really focused on SBCL, but patches to add support for other systems are more than welcome.

This library started out as a single file, `utilities.lisp`, that I personally used for shared generic code for all of my Lisp code. Most lispers have a similar file of some name, `utilities.lisp`, `misc.lisp`, `shared.lisp`, or even `stuff.lisp`, that is just a random collection of useful little generic macros and functions. Mine has grown over the years, and in 2012 I decided that I should try to make it useful to people other than myself.

You can download the library from GitHub at:

<https://github.com/cgore/sigma>

and I have some other information on it at my own website at:

<http://cgore.com/programming/lisp/sigma/>

## Getting Lisp

Before using this library you need a working Lisp. I use and recommend SBCL, Steel Bank Common Lisp, which is available at:

<http://www.sbcl.org>

This is derived from CMUCL, Carnegie Mellon University Common Lisp, which is still under active development and is available at:

<http://www.cons.org/cmucl/>

SBCL has information on getting started at:

<http://www.sbcl.org/getting.html>

If you are using Debian or a similar Linux distribution (including Ubuntu), you can just run as root:

```
apt-get install sbcl sbcl-doc sbcl-source
```

## Getting EMACS and SLIME

After installing, the best way to interact with any Common Lisp is via SLIME, the Superior Lisp Interaction Mode for EMACS, which is avail-

able at:

`http://common-lisp.net/project/slime/`

This can be installed on Debian by:

```
apt-get install slime emacs emacs-goodies-el
```

## Using the Library

First we need to clone the utilities.

```
mkdir -p /programming/lisp
cd /programming/lisp
git clone git@github.com:cgore/sigma.git
```

Now we need to make a directory for our project and symlink to the ASDF definition. There are other ways to load ASDF libraries, especially if you want to have them available globally; I strongly recommend you read the documentation to ASDF.

```
mkdir our-new-project
cd our-new-project
ln -s /programming/lisp/sigma/sigma.asd
```

Now we need to start up our Lisp REPL. The best way to do this for personal use is SLIME from within Emacs, but I will demonstrate using the shell itself here.

```
sbcl
```

Now we are in SBCL. Let's calculate something.

$$\sum_{i=0}^{100} i$$

```
(asdf:load-system :sigma) ; Load the Sigma system via ASDF.
(sigma:use-all-sigma) ; This will pollute COMMON-LISP-USER.
(sum (loop for i from 1 to 100
          collect i)) ; Returns 5050, and makes Euler sad.
```

Have fun!

# Chapter 1

## The `sigma/behave` Package

The `sigma/behave` package contains some useful code for confirming behavior of code, supporting a very basic form of *behavior-driven development*, BDD. The basic flow is to define the *behavior* of something, with multiple *specs* specified within that behavior specification, each consisting of various assertions, such as `should=`, `should-equal`, `should-not-equal`, and many others. If the behavior of the thing doesn't match the specified behavior, then there is some error.

### 1.1 Macros

#### 1.1.1 The behavior Macro

The `behavior` macro is used to specify a block of expected behavior for a `thing`. It specifies an example group, loosely similar to the `describe` blocks in Ruby's `RSpec`. It takes a single argument, the `thing` we are trying to describe, and then a body of code to evaluate that is evaluated in an implicit `progn`. It is to be used around a set of examples, or around a set of assertions directly.

#### Syntax

```
(behavior thing &body body)
```

#### Arguments and Values

***thing*** This is what we are describing the behavior of.

***body*** This is an implicit `progn` to contain the behavior.

## Examples

```
(behavior 'float
  (spec "is_an_Abelian_group"
    (let ((a (random 10.0))
          (b (random 10.0))
          (c (random 10.0))
          (e 1.0))
      (spec "closure"
        (should-be-a 'float (* a b)))
      (spec "associativity"
        (should= (* (* a b) c)
                  (* a (* b c))))
      (spec "identity_element"
        (should= a (* e a)))
      (spec "inverse_element"
        (let ((1/a (/ 1 a)))
          (should= (* 1/a a)
                    (* a 1/a)
                    1.0)))
      (spec "commutativity"
        (should= (* a b) (* b a))))))
```

### 1.1.2 The spec Macro

The `spec` macro is used to indicate a specification for a desired behavior. It will normally serve as a grouping for assertions or nested specs.

#### Syntax

```
(spec description &body body)
```

#### Arguments and Values

***description*** This is a string to describe the specification.

***body*** This is an implicit progn to contain the specification.

## Examples

```
(spec "should_pass_some_tests"
  (should= 12 (foo 3.5))
  (should= 14 (foo 4.22)))
```



### 1.1.3 The `should` Macro

The `should` macro is the basic building block for most of the behavior checking. It asserts that `test` returns truthfully for the arguments. Typically you will want to use one of the macros defined on top of `should` instead of using it directly, such as `should=`.

#### Syntax

```
(should test &rest arguments)
```

#### Arguments and Values

**test** This is the test predicate to evaluate.

**arguments** These are the arguments to the test predicate.

#### Examples

```
(should #'= 12 (* 3 4)) ; Passes  
(should #'< 4 (* 2 3)) ; Passes  
(should #'< 4 5 6 7)   ; Passes
```

### 1.1.4 The `should-not` Macro

The `should-not` macro is identical to the `should` macro, except that it inverts the result of the call with `not`.

#### Syntax

```
(should-not test &rest arguments)
```

#### Arguments and Values

**test** This is the test predicate to evaluate.

**arguments** These are the arguments to the test predicate.

#### Examples

```
(should-not #'< 12 4) ; Passes  
(should-not #'= 12 44) ; Passes
```

### 1.1.5 The should-be-null Macro

The `should-be-null` macro is a short-hand method for `(should #'null ...)`.

#### Syntax

```
(should-be-null &rest arguments)
```

#### Arguments and Values

**arguments** These are the arguments to `null`.

#### Examples

```
(should-be-null ()) ; Passes  
(should-be-null nil) ; Passes  
(should-be-null (not 12)) ; Passes  
(should-be-null (and t t nil)) ; Passes
```

### 1.1.6 The should-be-true Macro

The `should-be-true` macro is a short-hand method for `(should #'identity ...)`.

#### Syntax

```
(should-be-true &rest arguments)
```

#### Arguments and Values

**arguments** These are the arguments to `identity`.

#### Examples

```
(should-be-true t) ; Passes  
(should-be-true (not nil)) ; Passes  
(should-be-true (or nil nil 12)) ; Passes
```

### 1.1.7 The should-be-false Macro

The `should-be-false` macro is a short-hand method for `(should #'not ...)`.

#### Syntax

```
(should-be-false &rest arguments)
```

## Arguments and Values

**arguments** These are the arguments to `not`.

## Examples

```
(should-be-false nil)
(should-be-false (not t))
(should-be-false (< 44 2))
```

### 1.1.8 The `should-not-be-true` Macro

The `should-not-be-true` macro is a short-hand method for `(should-not #'identity ...)`. It asserts that the body does not evaluate to a true value.

## Syntax

```
(should-not-be-true &body body)
```

## Examples

```
(should-not-be-true nil)           ; Passes
(should-not-be-true (not t))       ; Passes
(should-not-be-true t)             ; Fails
```

### 1.1.9 The `should-not-be-false` Macro

The `should-not-be-false` macro is a short-hand method for `(should-not #'not ...)`. It asserts that the body does not evaluate to a false value; that is, that it is true.

## Syntax

```
(should-not-be-false &body body)
```

## Examples

```
(should-not-be-false t)             ; Passes
(should-not-be-false (not nil))     ; Passes
(should-not-be-false nil)           ; Fails
```

### 1.1.10 The `should-be-a` Macro

The `should-be-a` macro specifies that one or more things should be of the type specified by `type`.

**Syntax**

`(should-be-a type &rest things)`

**Arguments and Values**

**type** This is the type to compare with via `typep`.

**things** These are the things to confirm the type of.

```
(should-be-a 'integer 1)           ; Passes
(should-be-a 'float 1.0)          ; Passes
(should-be-a 'integer 1 2 3 4 5 6 7 8 9) ; Passes
(should-be-a 'integer 1.0)        ; Fails
```

**1.1.11 The should= Macro**

The `should=` macro is a short-hand method for `(should #'= ...)`.

**Syntax**

`(should= &rest arguments)`

**Arguments and Values**

**arguments** These are the arguments to `=`.

**Examples**

```
(should= 12 12)      ; Passes
(should= 12 12.0)    ; Passes
```

**1.1.12 The should-not= Macro**

The `should-not=` macro is a short-hand method for `(should-not #'= ...)`.

**Syntax**

`(should-not= &rest arguments)`

**Arguments and Values**

**arguments** These are the arguments to `=`.

**Examples**

```
(should-not= 12 12)    ; Fails  
(should-not= 12 12.0) ; Fails  
(should-not= 12 14)   ; Passes
```

**1.1.13 The should/= Macro**

The `should/=` macro is a short-hand method for `(should #'/= ...)`.

**Syntax**

```
(should/= &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to `/=`.

**Examples**

```
(should/= 12 13)    ; Passes  
(should/= 12 12)    ; Fails  
(should/= 12 12.0)  ; Fails
```

**1.1.14 The should-not/= Macro**

The `should-not/=` macro is a short-hand method for `(should-not #'/= ...)`.

**Syntax**

```
(should-not/= &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to `/=`.

**Examples**

```
(should-not/= 12 13)    ; Fails  
(should-not/= 12 12)    ; Passes  
(should-not/= 12 12.0)  ; Passes
```

### 1.1.15 The `should<` Macro

The `should<` macro is a short-hand method for `(should #'< ...)`.

#### Syntax

```
(should< &rest arguments)
```

#### Arguments and Values

**arguments** These are the arguments to `<`.

#### Examples

```
(should< 12 13) ; Passes  
(should< 13 12) ; Fails  
(should< 12 12) ; Fails
```

### 1.1.16 The `should-not<` Macro

The `should-not<` macro is a short-hand method for `(should-not #'< ...)`.

#### Syntax

```
(should-not< &rest arguments)
```

#### Arguments and Values

**arguments** These are the arguments to `<`.

#### Examples

```
(should-not< 12 13) ; Passes  
(should-not< 13 12) ; Fails  
(should-not< 12 12) ; Fails
```

### 1.1.17 The `should>` Macro

The `should>` macro is a short-hand method for `(should #'> ...)`.

#### Syntax

```
(should> &rest arguments)
```

**Arguments and Values**

*arguments* These are the arguments to >.

**Examples**

```
(should> 12 13) ; Fails  
(should> 13 12) ; Passes  
(should> 12 12) ; Fails
```

**1.1.18 The should-not> Macro**

The `should-not>` macro is a short-hand method for `(should-not #'> ...)`.

**Syntax**

```
(should-not> &rest arguments)
```

**Arguments and Values**

*arguments* These are the arguments to >.

**Examples**

```
(should-not> 12 13) ; Passes  
(should-not> 13 12) ; Fails  
(should-not> 12 12) ; Passes
```

**1.1.19 The should<= Macro**

The `should<=` macro is a short-hand method for `(should #'<= ...)`.

**Syntax**

```
(should<= &rest arguments)
```

**Arguments and Values**

*arguments* These are the arguments to <=.

### Examples

```
(should<= 12 13) ; Passes  
(should<= 13 12) ; Fails  
(should<= 12 12) ; Passes
```

#### 1.1.20 The should-not<= Macro

The `should-not<=` macro is a short-hand method for `(should-not #'<= ...)`.

### Syntax

```
(should-not<= &rest arguments)
```

### Arguments and Values

**arguments** These are the arguments to `<=`.

### Examples

```
(should-not<= 12 13) ; Fails  
(should-not<= 13 12) ; Passes  
(should-not<= 12 12) ; Fails
```

#### 1.1.21 The should>= Macro

The `should>=` macro is a short-hand method for `(should #'>= ...)`.

### Syntax

```
(should>= &rest arguments)
```

### Arguments and Values

**arguments** These are the arguments to `>=`.

### Examples

```
(should>= 12 13) ; Fails  
(should>= 13 12) ; Passes  
(should>= 12 12) ; Passes
```



### 1.1.22 The `should-not>=` Macro

The `should-not>=` macro is a short-hand method for `(should-not #'>= ...)`.

#### Syntax

```
(should-not>= &rest arguments)
```

#### Arguments and Values

**arguments** These are the arguments to `>=`.

#### Examples

```
(should-not>= 12 13) ; Passes  
(should-not>= 13 12) ; Fails  
(should-not>= 12 12) ; Fails
```

### 1.1.23 The `should-eq` Macro

The `should-eq` macro is a short-hand method for `(should #'eq ...)`.

#### Syntax

```
(should-eq &rest arguments)
```

#### Arguments and Values

**arguments** These are the arguments to `eq`.

#### Examples

```
(should-eq 12 12) ; Probably passes  
(should-eq 13 12) ; Fails  
(should-eq "foo" "foo") ; May pass, may fail.
```

### 1.1.24 The `should-not-eq` Macro

The `should-not-eq` macro is a short-hand method for `(should-not #'eq ...)`.

#### Syntax

```
(should-not-eq &rest arguments)
```

### Arguments and Values

**arguments** These are the arguments to `eq`.

### Examples

```
(should-not-eq 12 12)      ; Probably fails
(should-not-eq 13 12)      ; Passes
(should-not-eq "foo" "foo") ; May pass, may fail.
```

#### 1.1.25 The `should-eql` Macro

The `should-eql` macro is a short-hand method for `(should #'eql ...)`.

### Syntax

```
(should-eql &rest arguments)
```

### Arguments and Values

**arguments** These are the arguments to `eql`.

### Examples

```
(should-eql 12 12)      ; Passes
(should-eql 13 12)      ; Fails
(should-eql "foo" "foo") ; May pass, may fail.
```

#### 1.1.26 The `should-not-eql` Macro

The `should-not-eql` macro is a short-hand method for `(should-not #'eql ...)`.

### Syntax

```
(should-not-eql &rest arguments)
```

### Arguments and Values

**arguments** These are the arguments to `eql`.

**Examples**

```
(should-not-eql 12 12)           ; Fails  
(should-not-eql 13 12)           ; Passes  
(should-not-eql "foo" "foo") ; May pass, may fail.
```

**1.1.27 The should-equal Macro**

The `should-equal` macro is a short-hand method for `(should #'equal ...)`.

**Syntax**

```
(should-equal &rest arguments)
```

**Arguments and Values**

***arguments*** These are the arguments to `equal`.

**Examples**

```
(should-equal 12 12)           ; Passes  
(should-equal 13 12)           ; Fails  
(should-equal "foo" "foo") ; Passes  
(should-equal "FOO" "foo") ; Fails
```

**1.1.28 The should-not-equal Macro**

The `should-not-equal` macro is a short-hand method for `(should-not #'equal ...)`.

**Syntax**

```
(should-not-equal &rest arguments)
```

**Arguments and Values**

***arguments*** These are the arguments to `equal`.

**Examples**

```
(should-not-equal 12 12)           ; Passes  
(should-not-equal 13 12)           ; Fails  
(should-not-equal "foo" "foo") ; Fails  
(should-not-equal "FOO" "foo") ; Passes
```

### 1.1.29 The `should-equalp` Macro

The `should-equalp` macro is a short-hand method for `(should #'equalp ...)`.

#### Syntax

```
(should-equalp &rest arguments)
```

#### Arguments and Values

**arguments** These are the arguments to `equalp`.

#### Examples

```
(should-equalp 12 12)           ; Passes  
(should-equalp 13 12)           ; Fails  
(should-equalp "foo" "foo")     ; Passes  
(should-equalp "FOO" "foo")     ; Passes
```

### 1.1.30 The `should-not-equalp` Macro

The `should-not-equalp` macro is a short-hand method for `(should-not #'equalp ...)`.

#### Syntax

```
(should-not-equalp &rest arguments)
```

#### Arguments and Values

**arguments** These are the arguments to `equalp`.

#### Examples

```
(should-not-equalp 12 12)        ; Passes  
(should-not-equalp 13 12)        ; Fails  
(should-not-equalp "foo" "foo")  ; Passes  
(should-not-equalp "FOO" "foo")  ; Fails
```

### 1.1.31 The `should-string=` Macro

The `should-string=` macro is a short-hand method for `(should #'string= ...)`.

**Syntax**

```
(should-string= &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to `string=`.

**Examples**

```
(should-string= "foo" "foo") ; Passes  
(should-string= "FOO" "foo") ; Fails
```

**1.1.32 The should-not-string= Macro**

The `should-not-string=` macro is a short-hand method for `(should-not #'string= ...)`.

**Syntax**

```
(should-not-string= &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to `string=`.

**Examples**

```
(should-not-string= "foo" "foo") ; Fails  
(should-not-string= "FOO" "foo") ; Passes
```

**1.1.33 The should-string/= Macro**

The `should-string/=` macro is a short-hand method for `(should #'string/= ...)`.

**Syntax**

```
(should-string/= &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to `string/=`.

## Examples

```
(should-string/= "foo" "foo") ; Fails  
(should-string/= "FOO" "foo") ; Passes
```

### 1.1.34 The should-not-string/= Macro

The `should-not-string/=` macro is a short-hand method for `(should-not #'string/= ...)`.

#### Syntax

```
(should-not-string/= &rest arguments)
```

#### Arguments and Values

**arguments** These are the arguments to `string/=`.

## Examples

```
(should-not-string/= "foo" "foo") ; Passes  
(should-not-string/= "FOO" "foo") ; Fails
```

### 1.1.35 The should-string< Macro

The `should-string<` macro is a short-hand method for `(should #'string< ...)`.

#### Syntax

```
(should-string< &rest arguments)
```

#### Arguments and Values

**arguments** These are the arguments to `string<`.

## Examples

```
(should-string< "foo" "f") ; Fails  
(should-string< "foo" "foo") ; Fails  
(should-string< "foo" "FOOBAR") ; Fails  
(should-string< "foo" "foobar") ; Passes
```

### 1.1.36 The `should-not-string<` Macro

The `should-not-string<` macro is a short-hand method for `(should-not #'string< ...)`.

#### Syntax

```
(should-not-string< &rest arguments)
```

#### Arguments and Values

**arguments** These are the arguments to `string<`.

#### Examples

```
(should-not-string< "foo" "f")           ; Passes  
(should-not-string< "foo" "foo")         ; Passes  
(should-not-string< "foo" "foobar")      ; Fails
```

### 1.1.37 The `should-string>` Macro

The `should-string>` macro is a short-hand method for `(should #'string> ...)`.

#### Syntax

```
(should-string> &rest arguments)
```

#### Arguments and Values

**arguments** These are the arguments to `string>`.

#### Examples

```
(should-string> "foo" "f")               ; Passes  
(should-string> "foo" "foo")             ; Fails  
(should-string> "foo" "FOO")             ; Passes  
(should-string> "foo" "foobar")          ; Fails
```

### 1.1.38 The `should-not-string>` Macro

The `should-not-string>` macro is a short-hand method for `(should-not #'string> ...)`.

**Syntax**

```
(should-not-string> &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to `string>`.

**Examples**

```
(should-not-string> "foo" "f")           ; Fails  
(should-not-string> "foo" "foo")        ; Passes  
(should-not-string> "foo" "foobar")     ; Passes
```

**1.1.39 The should-string<= Macro**

The `should-string<=` macro is a short-hand method for `(should #'string<= ...)`.

**Syntax**

```
(should-string<= &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to `string<=`.

**Examples**

```
(should-string<= "foo" "f")              ; Fails  
(should-string<= "foo" "foo")            ; Passes  
(should-string<= "foo" "foobar")         ; Passes
```

**1.1.40 The should-not-string<= Macro**

The `should-not-string<=` macro is a short-hand method for `(should-not #'string<= ..`

**Syntax**

```
(should-not-string<= &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to `string<=`.



**Examples**

```
(should-not-string<= "foo" "f")           ; Passes  
(should-not-string<= "foo" "foo")        ; Fails  
(should-not-string<= "foo" "foobar")     ; Fails
```

**1.1.41 The should-string>= Macro**

The `should-string>=` macro is a short-hand method for `(should #'string>= ...)`.

**Syntax**

```
(should-string>= &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to `string>=`.

**Examples**

```
(should-string>= "foo" "f")               ; Passes  
(should-string>= "foo" "foo")             ; Passes  
(should-string>= "foo" "foobar")          ; Fails
```

**1.1.42 The should-not-string>= Macro**

The `should-not-string>=` macro is a short-hand method for `(should-not #'string>= ...)`.

**Syntax**

```
(should-not-string>= &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to `string>=`.

**Examples**

```
(should-not-string>= "foo" "f")           ; Fails  
(should-not-string>= "foo" "foo")         ; Fails  
(should-not-string>= "foo" "foobar")     ; Passes
```

### 1.1.43 The `should-string-equal` Macro

The `should-string-equal` macro is a short-hand method for `(should #'string-equal ...)`.

#### Syntax

```
(should-string-equal &rest arguments)
```

#### Arguments and Values

**arguments** These are the arguments to `string-equal`.

#### Examples

```
(should-string-equal "foo" "foo")      ; Passes  
(should-string-equal "FOO" "foo")     ; Passes  
(should-string-equal "foo" "foobar")  ; Fails
```

### 1.1.44 The `should-not-string-equal` Macro

The `should-not-string-equal` macro is a short-hand method for `(should-not #'string-equal ...)`.

#### Syntax

```
(should-not-string-equal &rest arguments)
```

#### Arguments and Values

**arguments** These are the arguments to `string-equal`.

#### Examples

```
(should-not-string-equal "foo" "foo")  ; Fails  
(should-not-string-equal "FOO" "foo")  ; Fails  
(should-not-string-equal "foo" "foobar") ; Passes
```

### 1.1.45 The `should-string-not-equal` Macro

The `should-string-not-equal` macro is a short-hand method for `(should #'string-not-equal ...)`.

#### Syntax

```
(should-string-not-equal &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to `string-not-equal`.

**Examples**

```
(should-string-not-equal "foo" "foo")      ; Fails  
(should-string-not-equal "FOO" "foo")     ; Fails  
(should-string-not-equal "foo" "foobar") ; Passes
```

**1.1.46 The should-not-string-not-equal Macro**

The `should-not-string-not-equal` macro is a short-hand method for `(should-not #'string-not-equal ...)`.

**Syntax**

```
(should-not-string-not-equal &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to `string-not-equal`.

**Examples**

```
(should-not-string-not-equal "foo" "foo")      ; Passes  
(should-not-string-not-equal "FOO" "foo")     ; Passes  
(should-not-string-not-equal "foo" "foobar") ; Fails
```

**1.1.47 The should-string-lessp Macro**

The `should-string-lessp` macro is a short-hand method for `(should #'string-lessp ...)`.

**Syntax**

```
(should-string-lessp &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to `string-lessp`.

**Examples**

```
(should-string-lessp "foo" "f")           ; Fails
(should-string-lessp "foo" "foo")         ; Fails
(should-string-lessp "foo" "FOOBAR")      ; Passes
(should-string-lessp "foo" "foobar")      ; Passes
```

**1.1.48 The should-not-string-lessp Macro**

The `should-not-string-lessp` macro is a short-hand method for `(should-not #'string-lessp ...)`.

**Syntax**

```
(should-not-string-lessp &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to `string-lessp`.

**Examples**

```
(should-not-string-lessp "foo" "f")       ; Passes
(should-not-string-lessp "foo" "foo")     ; Passes
(should-not-string-lessp "foo" "FOOBAR")  ; Fails
(should-not-string-lessp "foo" "foobar")  ; Fails
```

**1.1.49 The should-string-greaterp Macro**

The `should-string-greaterp` macro is a short-hand method for `(should #'string-greaterp ...)`.

**Syntax**

```
(should-string-greaterp &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to `string-greaterp`.

**Examples**

```
(should-string-greaterp "foo" "f")        ; Passes
(should-string-greaterp "foo" "foo")      ; Fails
(should-string-greaterp "foo" "FOO")      ; Fails
(should-string-greaterp "foo" "foobar")   ; Fails
```

### 1.1.50 The `should-not-string-greaterp` Macro

The `should-not-string-greaterp` macro is a short-hand method for `(should-not #'string-greaterp ...)`.

#### Syntax

```
(should-not-string-greaterp &rest arguments)
```

#### Arguments and Values

**arguments** These are the arguments to `string-greaterp`.

#### Examples

```
(should-not-string-greaterp "foo" "f")           ; Fails  
(should-not-string-greaterp "foo" "foo")         ; Passes  
(should-not-string-greaterp "foo" "FOO")         ; Passes  
(should-not-string-greaterp "foo" "foobar")      ; Passes
```

### 1.1.51 The `should-string-not-greaterp` Macro

The `should-string-not-greaterp` macro is a short-hand method for `(should #'string-not-greaterp ...)`.

#### Syntax

```
(should-string-not-greaterp &rest arguments)
```

#### Arguments and Values

**arguments** These are the arguments to `string-not-greaterp`.

#### Examples

```
(should-string-not-greaterp "foo" "f")           ; Fails  
(should-string-not-greaterp "foo" "foo")         ; Passes  
(should-string-not-greaterp "foo" "FOO")         ; Passes  
(should-string-not-greaterp "foo" "foobar")      ; Passes
```

### 1.1.52 The `should-not-string-not-greaterp` Macro

The `should-not-string-not-greaterp` macro is a short-hand method for `(should-not #'string-not-greaterp ...)`.

**Syntax**

```
(should-not-string-not-greaterp &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to string-not-greaterp.

**Examples**

```
(should-not-string-not-greaterp "foo" "f")           ; Passes  
(should-not-string-not-greaterp "foo" "foo")        ; Fails  
(should-not-string-not-greaterp "foo" "FOO")        ; Fails  
(should-not-string-not-greaterp "foo" "foobar")     ; Fails
```

**1.1.53 The should-string-not-lessp Macro**

The should-string-not-lessp macro is a short-hand method for (should #'string-not-lessp ...).

**Syntax**

```
(should-string-not-lessp &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to string-not-lessp.

**Examples**

```
(should-string-not-lessp "foo" "f")                 ; Passes  
(should-string-not-lessp "foo" "foo")               ; Passes  
(should-string-not-lessp "foo" "FOOBAR")            ; Fails  
(should-string-not-lessp "foo" "foobar")            ; Fails
```

**1.1.54 The should-not-string-not-lessp Macro**

The should-not-string-not-lessp macro is a short-hand method for (should-not #'string-not-lessp ...).

**Syntax**

```
(should-not-string-not-lessp &rest arguments)
```

**Arguments and Values**

**arguments** These are the arguments to `string-not-lessp`.

**Examples**

```
(should-not-string-not-lessp "foo" "f")           ; Fails  
(should-not-string-not-lessp "foo" "foo")         ; Fails  
(should-not-string-not-lessp "foo" "FOOBAR")      ; Passes  
(should-not-string-not-lessp "foo" "foobar")      ; Passes
```





## Chapter 2

# The `sigma/control` Package

The `sigma/control` package contains code for basic program control systems. These are mostly basic macros to add more complicated looping, conditionals, or similar. These are typically extensions to Common Lisp that are inspired by other programming languages. Thanks to the power of Common Lisp and its macro system, we can typically implement most features of any other language with little trouble.

### 2.1 Macros

#### 2.1.1 The `aif` Macro

The `aif` macro is an anaphoric variation of the built-in `if` control structure. This is based on [1, p. 190]. The basic idea is to provide an anaphor (such as pronouns in English) for the conditional so that it can easily be referred to within the body of the conditional expression. The most natural pronoun in the English language for a thing is “it”, so that is what is used. If you need or want to use a different anaphor, use `a?if`. The most common use of `aif` is for when you want to do some additional computation with some time-consuming calculation, but only if it returned successfully.

#### Syntax

```
(aif conditional t-action &optional nil-action)
```

**Arguments and Values**

**conditional** The boolean conditional to select between the *t-action* and the *nil-action*.

**t-action** The action to evaluate if the *conditional* evaluates as true.

**nil-action** The action to evaluate if the *conditional* evaluates as nil.

**Examples**

```
(aif (big-long-calculation)
     (foo it)
     (format t "The_big-long-calculation_failed!~%"))
```

This is similar to the following, but with less typing:

```
(let ((it (big-long-calculation)))
  (if it
      (foo it)
      (format t "The_big-long-calculation_failed!~%")))
```

Or say you need to get a user name from a database call, which might be slow.

```
(aif (get-user-name)
     (format -t "Hello, ~A!~%" it)
     (format -t "You_aren't_logged_in,_go_away!~%"))
```

**2.1.2 The a?if Macro**

The *a?if* macro is a variation of *aif* that allows for the specification of the anaphor to use, instead of being restricted to just *it*, the default with *aif*. This is most often useful when you need to nest calls to anaphoric macros.

**Syntax**

```
(a?if anaphor conditional t-action &optional nil-action)
```

**Arguments and Values**

**anaphor** The result of the *conditional* will be stored in the variable specified as the anaphor.

**conditional** The boolean conditional to select between the *t-action* and the *nil-action*.

**t-action** The action to evaluate if the *conditional* evaluates as true.

***nil-action*** The action to evaluate if the *conditional* evaluates as nil.

### Examples

```
(a?if foo 'outer
  (a?if bar 'inner
    `(:,foo ,bar))) ; Returns '(outer inner)
```

### 2.1.3 The aand Macro

The `aand` macro is an anaphoric variation of the built-in `and`. This is based on [1, p. 191]. It works in a similar manner to `aif`, defining `it` as the current argument for use in the next argument, reassigning `it` with each argument.

### Syntax

```
(aand &rest arguments)
```

### Examples

```
(aand 2          ; Sets 'it' to 2.
  (* 3 it)      ; Sets 'it' to 6.
  (* 4 it))    ; Returns 24.
```

### 2.1.4 The a?and Macro

The `a?and` macro is a variant of `aand` that allows for the specification of the anaphor to use, instead of being restricted to just `it`, the default with `aand`. This is most often useful when you need to nest calls to anaphoric macros.

### Examples

```
(a?and foo 12 (* 2 foo) (* 3 foo)) ; Returns 72.

(a?and foo 1 2 3 'outer
  (a?and bar 4 5 6 'inner `(:,foo ,bar))) ; Returns '(outer inner)
```

### 2.1.5 The `alambda` Macro

The `alambda` macro is an anaphoric variant of the built-in `lambda`. This is based on [1, p. 193]. It works in a similar manner to `aif` and `aand`, except it defines `self` instead of `it` as the default anaphor. This is useful so that you can write recursive lambdas.

```
(funcall (alambda (x) ; Simple recursive factorial example.
  (if (<= x 0)
    1
    (* x (self (1- x)))))
10))) ; Calculates 10!, inefficiently.
```

### 2.1.6 The `a?lambda` Macro

The `a?lambda` macro is a variant of `alambda` that allows you to specify the anaphor to use, instead of just the default of `it`.

```
(funcall (a?lambda ! (x) ; Simple recursive factorial example.
  (if (<= x 0)
    1
    (* x (! (1- x)))))
10))) ; Calculates 10!, inefficiently.
```

### 2.1.7 The `ablock` Macro

The `ablock` macro is an anaphoric variant of the built-in `block`. This is based on [1, p. 193]. It works in a similar manner to `aand`, defining the anaphor `it` for each argument to the block.

#### Examples

```
(let (w x y z)
  (ablock b
    (setf w 7)
    (setf x (* 2 it)) ; Twice w, 14.
    (setf y (* 3 it)) ; Thrice x, 42.
    (return-from b)   ; Leave the block.
    (setf z 123))     ; Never happens.
  (list w x y z))     ; Returns '(7 14 42 nil)
```

### 2.1.8 The `a?block` Macro

The `a?block` macro is an anaphoric variant of `ablock` that allows you to specify the anaphor to use, instead of just the default of `it`.

**Examples**

```
(let (w x y z)
  (a?block b foo
    (setf w 7)
    (setf x (* 2 foo)) ; Twice w, 14.
    (setf y (* 3 foo)) ; Thrice x, 42.
    (return-from b)    ; Leave the block.
    (setf z 123))      ; Never happens.
  (list w x y z))      ; Returns '(7 14 42 nil)
```

**2.1.9 The acond Macro**

The `acond` macro is an anaphoric variant of the built-in `cond`. This is based on [1, p. 191]. It works in a similar manner to `aand`, defining the anaphor `it` for each argument to the conditional.

**Examples**

```
(let (a b (c 3))
  (acond (a it)          ; No.
        (b it)          ; No.
        (c (* 4 it)))) ; Yes, returns 12 = 4*3, the value of c.
```

**2.1.10 The a?cond Macro**

The `a?cond` macro is an anaphoric variant of `acond` that allows you to specify the anaphor to use, instead of just the default of `it`.

**Examples**

```
(let (a b (c 3))
  (a?cond foo
    (a foo)          ; No.
    (b foo)          ; No.
    (c (* 4 foo)))) ; Yes, returns 12 = 4*3, the value of c.
```

**2.1.11 The awhen Macro**

The `awhen` macro is an anaphoric variant of the built-in `when`. This is based on [1, p. 191]. It works in a similar manner to `aif`, defining `it` as the default anaphor. This is useful when the conditional is the result of a complicated computation, so you don't have to compute it twice or wrap the computation in a `let` block yourself.

**Syntax**

```
(awhen conditional &body body)
```

**Examples**

```
(awhen (get-user-name)
  (do-something-with-name it)
  (do-more-stuff)
  (format -t "Hello, ~A!~%" it))
```

**2.1.12 The a?when Macro**

The `a?when` macro is similar to the `awhen`, except that it allows you to specify the anaphor to use, instead of just the default of `it`.

**Syntax**

```
(a?when anaphor conditional &body body)
```

**Examples**

```
(a?when user (get-user-name)
  (do-something-with-name user)
  (do-more-stuff)
  (format -t "Hello, ~A!~%" user))
```

**2.1.13 The awhile Macro**

The `awhile` macro is an anaphoric variant of `while`. This is based on [1, p. 191]. This is useful if you need to consume input repeatedly for all input. It returns the value of the last form in the body from the last iteration, or `nil` if the body never runs.

**Syntax**

```
(awhile expression &body body)
```

**Examples**

```
(awhile (get-input)
  (do-something it)) ; Operate on input for all input.

(let ((forward '(1 2 3))
      (backward nil))
```

```
(awhile (pop forward)
  (push it backward)) ; Returns (3 2 1)
```

### 2.1.14 The a?while Macro

The `a?while` macro is a variant of `awhile` that allows you to specify the anaphor to use, instead of just the default `it`. It returns the value of the last form in the body from the last iteration, or `nil` if the body never runs.

#### Syntax

```
(a?while anaphor expression &body body)
```

#### Examples

```
(a?while input (get-input)
  (do-something input)) ; Operate on input for all input.
```

### 2.1.15 The defconstant-once Macro

The `defconstant-once` macro defines a constant using `defconstant`, but only if it is not already bound. It expands to a top-level `defconstant` inside `eval-when` so the compiler can see the name when compiling later forms in the same file; on reload the value form returns the existing binding so the constant is not redefined inconsistently. This is particularly useful for large immutable datasets so that reloading a file with ASDF does not produce redefinition warnings.

#### Syntax

```
(defconstant-once name value &optional docstring)
```

#### Arguments and Values

**name** The name of the constant (conventionally using `+plus-signs+`).

**value** The value of the constant. This expression is evaluated only the first time the constant is defined.

**docstring** An optional documentation string for the constant.

**Examples**

```
(defconstant-once +reward-al-stats+
  '((697 46.467 1345840.30 0.49023)
    ...))
"Statistics_from_reward-al_runs.")
```

This macro is defined in the `sigma/control` package.

**2.1.16 The deletef Macro**

The `deletef` macro deletes item from sequence in-place.

**Syntax**

```
(deletef item sequence &rest rest)
```

**Examples**

```
(let ((men '(good bad ugly)))
  (deletef 'bad men)
  (deletef 'ugly men)
  men) ; Only the good is left.
```

**2.1.17 The do-while Macro**

The `do-while` macro operates like a `do {BODY} while (CONDITIONAL)` in the C programming language. The body always runs at least once. The macro returns the value of the last form in the body from the last iteration.

**Syntax**

```
(do-while conditional &body body)
```

**Examples**

```
(let ((t-minus 10))
  (do-while (<= 0 t-minus)
    (format t "~A..." t-minus)
    (decf t-minus)))
(format t "Liftoff!~%")
```



**2.1.18 The do-until Macro**

The `do-until` macro operates like a `do {body} while (! conditional)` in the C programming language. The body always runs at least once. The macro returns the value of the last form in the body from the last iteration.

**Syntax**

```
(do-until conditional &body body)
```

**Examples**

```
(let ((t-minus 10))
  (do-until (< t-minus 0)
    (format t "~A_..._" t-minus)
    (decf t-minus)))
(format t "Liftoff!~%")
```

**2.1.19 The fop Macro**

`fop` is like the `opf` macro, but as a post-assignment variant. The difference is similar to the difference between `x++` and `++x` in the C Programming Language, with `opf` being like `++x` and `fop` being like `x++`.

**Syntax**

```
(fop operator variable &rest arguments)
```

**Examples**

```
(let ((x 10))
  (while (<= 0 x)
    (format t "~A_..._" (fop #'- x 1))))
(format t "Liftoff!~%")
```

**2.1.20 The for Macro**

A `for` macro, much like the `for` in the C programming language.

**Syntax**

```
(for initial conditional step-action &body body)
```

**Examples**

```
(for ((i 0))
  (< i 10)
  (incf i)
  (format t "~%~A\"_i))
```

**2.1.21 The forever Macro**

The `forever` macro is just a way to say `(while t ...)` with a bit of added expressiveness and explicitness.

**Examples**

```
(forever (let ((in (read)))
  (if (eq in 'quit)
      (format t "I_can't_let_you_do_that,_Dave.")
      (format t "You_entered_~A" in))))
```

**2.1.22 The multicond Macro**

The `multicond` macro is much like `cond`, but where multiple clauses may be evaluated.

**Examples**

```
(let ((x 12))
  (multicond ((= x 12) ; This will evaluate.
              (format t "X_is_12!_My_favorite_number!~%"))
              (< x 100) ; This will evaluate also.
              (format t "X_is_small.~%"))
              (< x 0) ; But this one won't.
              (format t "X_is_negative.~%")))))
```

**2.1.23 The opf Macro**

The `opf` macro is a generic operate-and-store, along the lines of `incf` and `decf`, but allowing for any operation.

**Syntax**

```
(opf operator variable &rest arguments)
```

**Examples**

```
;;; Prints 1 ... 2 ... 4 ... 8 ... ... 65535 ... that's it!
(let ((x 1))
  (while (<= x (expt 2 16))
    (format t "~A_" x)
    (opf #'* x 2)))
(format t "_that's_it!~%")
```

**2.1.24 The swap Macro**

This is a simple `swap` macro. The values of the first and second form are swapped with each other.

**Syntax**

```
(swap x y)
```

**Examples**

```
(let ((first "the_first")
      (last "the_last"))
  (swap first last)
  `(:,first ,last)) ; Returns '("the last" "the first")
```

**2.1.25 The swap-unless Macro**

This macro calls `swap` unless the predicate evaluates to true.

**Syntax**

```
(swap-unless predicate x y)
```

**Examples**

```
;;; make smaller and larger in the correct order.
(let ((smaller 12)
      (larger 266))
  (swap-unless #'<= smaller larger))
```

**2.1.26 The swap-when Macro**

This macro calls `swap` when the predicate evaluates to true.

**Syntax**

```
(swap-when predicate x y)
```

**Examples**

```
;;; make smaller and larger in the correct order.  
(let ((smaller 12)  
      (larger 6))  
  (swap-when #'> smaller larger))
```

**2.1.27 The until Macro**

The `until` macro is similar to the `while` loop in C, but with a negated conditional. It returns the value of the last form in the body from the last iteration, or `nil` if the body never runs.

**Syntax**

```
(until conditional &body body)
```

**Examples**

```
(let ((x 10))  
  (until (< x 0)  
    (format t "~A_..._" x)  
    (decf x))  
  (format t "Liftoff!~%"))  
  
(let ((x 0))  
  (until (<= 10 x)  
    (incf x))) ; Returns 10
```

**2.1.28 The while Macro**

This `while` macro is similar to the `while` loop in C. It returns the value of the last form in the body from the last iteration, or `nil` if the body never runs.

**Syntax**

```
(while conditional &body body)
```

**Examples**

```
(let ((x 10))
  (while (<= 0 x)
    (format t "~A_..." x)
    (decf x))
  (format t "Liftoff!~%"))

(let ((x 0))
  (while (< x 3)
    (incf x)
    (* x 10))) ; Returns 30
```

**2.1.29 The macro-alias Macro**

The `macro-alias` macro produces one or more alternate names for a macro by copying its macro function.

**Syntax**

```
(macro-alias macro &rest aliases)
```

**Examples**

```
(macro-alias when whenever)
```

**2.2 Functions****2.2.1 The compose Function**

The `compose` function composes a single function from a list of several functions such that the new function is equivalent to calling the functions in succession. This is based upon a `compose` function in [2], which is based upon the `compose` function from Dylan.

**Syntax**

```
(compose &rest functions)
```

**Examples**

We want to calculate:

$$\sin(\cos(\tan(\pi))) \approx 0.841,470,984,807,896,5$$

```
(funcall (compose #'sin #'cos #'tan) pi)
(sin (cos (tan pi))) ; This is the same.
```

### 2.2.2 The `conjoin` Function

The `conjoin` function takes in one or more predicates, and returns a predicate that returns true whenever all of the predicates return true. This is from [2] and is based upon the `conjoin` function from Dylan.

#### Syntax

```
(conjoin predicate &rest predicates)
```

#### Examples

```
;;; Returns '(6 12 18 24 30 36 42 48 54 60 66 72 78 84 90 96).
(remove-if-not #'identity
  (flet ((mod-2? (i) (zerop (mod i 2)))
        (mod-3? (i) (zerop (mod i 3))))
    (loop for i from 1 to 100 collect
      (when (funcall (conjoin #'mod-2? #'mod-3?) i)
        i))))
```

### 2.2.3 The `curry` Function

The `curry` function takes in a function and some of its arguments, and returns a function that expects the rest of the required arguments. This is from [2] and is based upon the `curry` function from Dylan.

#### Syntax

```
(curry function &rest arguments)
```

#### Examples

```
(let ((x 100)
      (f (curry #' + x)))
  (loop for i from 1 to 10 collect
    (funcall f i))) ; Returns '(101 102 103 ... 110)
```

### 2.2.4 The `disjoin` Function

The `disjoin` function takes in one or more predicates, and returns a predicate that returns true whenever any of the predicates return true. This is from [2] and is based upon the `disjoin` function from Dylan.

**Syntax**

```
(disjoin predicate &rest predicates)
```

**Examples**

```
;;; Returns ' (#\1 #\2 #\3 #\a #\b #\c NIL      NIL)
(let ((chars ' (#\1 #\2 #\3 #\a #\b #\c #\Space #\Newline)))
  (mapcar (lambda (c)
            (when (funcall (disjoin #'alpha-char-p #'digit-char-p)
                           c)
              c))
          chars))
```

**2.2.5 The function-alias Macro**

The `function-alias` macro produces one or more aliases (alternate names) for a function. Each name in `aliases` is given the same `fdefinition` as `function`.

**Syntax**

```
(function-alias function &rest aliases)
```

**Examples**

```
(function-alias 'that-guy-doesnt-know-when-to-stop-typing 'shorter)
(function-alias 'long-name 'short-a 'short-b)
```

**2.2.6 The function-aliases Macro**

The `function-aliases` macro is an alias for `function-alias`.

**2.2.7 The function-alias-as-a-function Function**

The `function-alias-as-a-function` function is the older functional form of `function-alias`. It assigns the function definition of `function` to each name in `aliases` by modifying `fdefinition`. Prefer the `function-alias` macro in new code.

**Syntax**

```
(function-alias-as-a-function function &rest aliases)
```

**Examples**

```
(function-alias-as-a-function 'long-name 'short)
```

**2.2.8 The function-aliases-as-a-function Function**

The `function-aliases-as-a-function` function is an alias for `function-alias-as-a-function`.

**2.2.9 The operator-to-function Function**

The `operator-to-function` function takes in any symbol and makes an evaluable function out of it. The principle purpose for this is so that we can treat macros and other non-function things like a function, for using them with `mapcar` or similar.

**Known Issues**

[Issue #8]

**Syntax**

```
(operator-to-function operator)
```

**Examples**

```
;;; In case you don't like (setf a 1 b 2 c 3).
(mapcar (operator-to-function 'setf)
        '(a b c)
        '(1 2 3))
```

**2.2.10 The rcompose Function**

The `rcompose` function is a reversed variant of the `compose` function.

**Syntax**

```
(rcompose &rest functions)
```

**Examples**

We want to calculate:

$$\tan(\cos(\sin(\pi))) \approx 1.557,407,724,654,902,3$$

```
(funcall (rcompose #'sin #'cos #'tan) pi)
(tan (cos (sin pi))) ; This is the same.
```



### 2.2.11 The `rcurry` Function

This function takes in a function and some of its ending arguments, and returns a function that expects the rest of the required arguments. This is from [2] and is based upon the `rcurry` function from Dylan.

#### Syntax

```
(rcurry function &rest arguments)
```

#### Examples

```
(let ((x 100)
      (f (rcurry #'- x)))
  (loop for i from 1 to 10 collect
        (funcall f i))) ; Returns '(-99 -98 -97 ... -90)
```

### 2.2.12 The `unimplemented` Function

This is a convenience function that merely raises an error. It is for code that is yet to be written.

#### Syntax

```
(unimplemented)
```

#### Examples

```
(defun turing-test-solver ()
  (unimplemented)) ; TODO: figure out how to program this.
```

## 2.3 Symbols

### 2.3.1 The `it` Variable

The symbol `it` is the default anaphor used by macros such as `aif`, `aand`, `awhen`, `awhile`, `acond`, and `ablock`. It is exported so that client code may refer to the anaphor without package qualification. It is not a global variable; each anaphoric macro binds it locally.

### 2.3.2 The self Variable

The symbol `self` is the default anaphor used by `alambda` for recursive calls. It is exported so that client code may refer to it without package qualification. It is not a global variable; `alambda` binds it locally as a local function.

## 2.4 Generics

### 2.4.1 The duplicate Generic

The `duplicate` generic is to provide a deep copy facility for any of your objects. If you define a class and want a deep copy facility for it, implement a version of `duplicate` that is correct for it. This library provides methods for lists (deeply mapped), arrays (deeply copied element-wise), numbers, symbols, and functions (the last three returned unchanged, being immutable or not meaningfully copyable).

#### Syntax

```
(duplicate item)
```

#### Examples

```
(duplicate '(1 2 (3 4))) ; Returns a fresh tree equal to the original  
(duplicate 42)          ; Returns 42
```

## Chapter 3

# The sigma/hash Package

### 3.1 Macros

#### 3.1.1 The `sethash` Macro

The `sethash` macro is a shortcut for `setf gethash`.

##### Syntax

```
(sethash key hash-table value)
```

##### Arguments and Values

***key*** The key to set in the hash table.

***hash-table*** The hash table we are modifying.

***value*** The value to store under that key.

##### Returns

The new *value* for the key.

#### 3.1.2 The `set-partition` Macro

The `set-partition` macro is a variant of `sethash` that assumes the hash entries are all sequences, allowing for multiple results per key. When you call `gethash` on the hashmap you will get back a sequence of all the entries for that key.

##### Syntax

```
(set-partition key hash-table value)
```

**Arguments and Values**

**key** The key to add to in the hash table.

**hash-table** The partition, a hash table, we are modifying.

**value** The value to add to the hash table key.

**Returns**

The new set of values for key, a sequence containing your newly added *value* and any previously added values.

## 3.2 Functions

### 3.2.1 The populate-hash-table Function

The `populate-hash-table` function makes initial construction of hash tables a lot easier, just taking in key/value pairs as the arguments to the function, and returning a newly-constructed hash table.

**Examples**

```
(populate-hash-table 'name "Valentinus"
                     'likes ' (birds roses)
                     'dislikes ' (beheadings epilepsy "false_idols")
                     'died 269)
```

### 3.2.2 The inchash Function

The `inchash` function will increment the value in *key* of the *hash*, initializing it to 1 if it isn't currently defined.

### 3.2.3 The dechash Function

The `dechash` function will decrement the value in *key* of the *hash*, initializing it to -1 if it isn't currently defined.

### 3.2.4 The gethash-in Function

The `gethash-in` function works like `gethash`, but allows for multiple keys to be specified at once, to work with nested hash tables.

**Syntax**

```
(gethash-in keys hash-table &optional default)
```

**Arguments and Values**

**keys** A list of objects.

**hash-table** A hash table.

**default** An object. The default is nil.

**Returns**

**value** An object.

**present?** A generalized boolean.

**Examples**

```
(let ((h (make-hash-table)))
  (sethash 'a h 12)
  (gethash-in '(a) h)) ; Returns 12

(let ((h (make-hash-table))
      (i (make-hash-table)))
  (sethash 'b i 123)
  (sethash 'a h i)
  (gethash-in '(a b) h 123)) ; Returns 123
```

**3.2.5 The make-partition Function**

The `make-partition` function is a variant of `make-hash-table` that assumes you're going to use the hash table for partitions with multiple entries per key, not a one-to-one hashmap.

**Syntax**

```
(make-partition)
```

**3.2.6 The populate-partition Function**

The `populate-partition` function makes initial construction of a partition a lot easier, just taking in key/value pairs as the arguments to the function, where there can be multiple entries for any key, and returning a newly-constructed partition.

**Syntax**

```
(populate-partition &rest pairs)
```

**Examples**

```
(populate-partition 'a ' (1 2 3)
  'a 4
  'a ' (5 6 7)
  'b 8
  'c 9
  'c 10
  'd 11
  'a 147)
```

## Chapter 4

# The sigma/numeric Package

### 4.1 Macros

#### 4.1.1 The `+f` Macro

The `+f` macro is an alias for `incf`.

#### 4.1.2 The `-f` Macro

The `-f` macro is an alias for `decf`.

#### 4.1.3 The `*f` Macro

The `*f` macro is an alias for `multf`.

#### 4.1.4 The `/f` Macro

The `/f` macro is an alias for `divf`.

#### 4.1.5 The `divf` Macro

The `divf` macro is divide-and-store, along the lines of `incf` and `decf`, but with division instead. This is similar to `x /= something` in the C programming language.

#### Syntax

`(divf variable &rest arguments)`

**Examples**

```
;;; Prints 65536 ... 8 ... 4 ... 2 ... 1 ... 0 ... that's it!
(let ((x (expt 2 16)))
  (while (<= 0 x)
    (format t "~A..." x)
    (divf x 2)))
(format t "that's it!~%")
```

**4.1.6 The f+ Macro**

The `f+` macro is similar to `incf` or `+f`, but it is a post-increment instead of a pre-increment. That is, `f+` works like `x++` in C but `incf` and `+f` work like `++x` in C.

**Syntax**

```
(f+ variable &rest addends)
```

**Examples**

```
(let ((x 12))
  (list x (f+ x) x)) ; Returns '(12 12 13).
```

**4.1.7 The f- Macro**

The `f-` macro is similar to `decf` or `-f`, but it is a post-decrement instead of a pre-decrement. That is, `f-` works like `x--` in C but `decf` and `-f` work like `--x` in C.

**Syntax**

```
(f- variable &rest subtrahends)
```

**Examples**

```
(let ((x 12))
  (list x (f- x) x)) ; Returns '(12 12 11).
```

**4.1.8 The f\* Macro**

The `f*` macro is similar to `multf` or `*f`, but it is a post-multiply instead of a pre-multiply. That is, `f*` works like `x++` in C (just for multiplication instead of addition) but `multf` and `*f` work like `++x` in C (again, just for multiplication instead of addition.)



**Syntax**

```
(f* variable &rest multiplicands)
```

**Examples**

```
(let ((x 12))
  (list x (f* x 2) x)) ; Returns '(12 12 24).
```

**4.1.9 The f/ Macro**

The `f/` macro is similar to `divf` or `/f`, but it is a post-divide instead of a pre-divide. That is, `f/` works like `x++` in C (just for division instead of addition) but `divf` and `/f` work like `++x` in C (again, just for division instead of addition.)

**Syntax**

```
(f/ variable &rest divisors)
```

**Examples**

```
(let ((x 12))
  (list x (f/ x 2) x)) ; Returns '(12 12 6).
```

**4.1.10 The multf Macro**

The `multf` macro is multiply-and-store, along the lines of `incf` and `decf`, but with multiplication instead. This is similar to `x *= something` in the C programming language.

**Syntax**

```
(multf variable &rest arguments)
```

**Examples**

```
;;; Prints 1 ... 2 ... 4 ... 8 ... ... 65535 ... that's it!
(let ((x 1))
  (while (<= x (expt 2 16))
    (format t "~A~..." x)
    (multf x 2)))
(format t "~that's it!~%")
```

## 4.2 Functions

### 4.2.1 The bit? Function

The `bit?` function returns true if its argument is of type `bit` (that is, either 0 or 1).

#### Syntax

```
(bit? b)
```

#### Examples

```
(bit? 0)    ; Returns T  
(bit? 1)    ; Returns T  
(bit? 2)    ; Returns NIL  
(bit? 0.5)  ; Returns NIL
```

### 4.2.2 The choose Function

The `choose` function computes the binomial coefficient for  $n$  and  $k$ , typically spoken as  $n$  choose  $k$ , and usually written mathematically as  $\binom{n}{k}$ .

#### Syntax

```
(choose n k)
```

#### Arguments and Values

$n$  A positive integer.

$k$  A positive integer such that  $0 < k < n$ .

#### Examples

```
(choose 12 2) ; Returns 66
```

### 4.2.3 The factorial Function

The `factorial` function computes  $n!$  for positive integers. NB, this is not intelligent, and uses a loop instead of better approaches.

#### Syntax

```
(factorial n)
```

**Arguments and Values**

*n* A positive integer.

**Examples**

```
(factorial 5) ; Returns 120
(factorial 1) ; Returns 1
```

**4.2.4 The fractional-part Function**

The `fractional-part` function returns the fractional part of a number in the form most familiar to computer scientists. It has the useful property that

$$\text{frac}(x) + \text{int}(x) = x,$$

but the result may be negative. See <http://mathworld.wolfram.com/FractionalPart.html>.

**Syntax**

```
(fractional-part number)
```

**Examples**

```
(fractional-part 10.5) ; Returns 0.5
(fractional-part 10)   ; Returns 0
(fractional-part -10.5) ; Returns -0.5
(fractional-part -10)  ; Returns 0
```

**4.2.5 The fractional-value Function**

The `fractional-value` function returns the fractional value of a number in the form most familiar to mathematicians. The result is always nonnegative, and forms a sawtooth wave. This is known as `SawtoothWave` in *Mathematica*. See <http://mathworld.wolfram.com/FractionalPart.html>. The symbol `sawtooth-wave` is an alias for this function.

**Syntax**

```
(fractional-value number)
(sawtooth-wave number)
```

**Examples**

```
(fractional-value 10.5) ; Returns 0.5
(fractional-value 10)   ; Returns 0
(fractional-value -10.5) ; Returns 0.5
(fractional-value -10)  ; Returns 0
```

**4.2.6 The integer-range Function**

The `integer-range` function generates lists of integer ranges of the form `[start, stop]`. It has three forms.

**Syntax**

```
(integer-range stop)
  (integer-range start stop)
  (integer-range start stop step)
```

**Arguments and Values**

**stop** The inclusive end of the range. When it is the sole argument, the range begins at 0.

**start** The inclusive start of the range.

**step** The stride between successive values. When omitted, the step is 1 or -1 according to the direction from start to stop.

**Examples**

```
(integer-range 10)
;; Returns '(0 1 2 3 4 5 6 7 8 9 10)

(integer-range 5 10)
;; Returns '(5 6 7 8 9 10)

(integer-range 5 10 2)
;; Returns '(5 7 9)

(integer-range -5 0)
;; Returns '(-5 -4 -3 -2 -1 0)

(integer-range 10 5 -1)
;; Returns '(10 9 8 7 6 5)
```

### 4.2.7 The nonnegative? Function

The `nonnegative?` function returns true if its argument is not negative (that is, if  $x \geq 0$ ).

#### Syntax

```
(nonnegative? x)
```

#### Examples

```
(nonnegative? 0)      ; Returns T  
(nonnegative? 3.5)   ; Returns T  
(nonnegative? -1)    ; Returns NIL
```

### 4.2.8 The nonnegative-integer? Function

The `nonnegative-integer?` function returns true if its argument is an integer greater than or equal to 0.

#### Syntax

```
(nonnegative-integer? x)
```

#### Examples

```
(nonnegative-integer? 12) ; Returns T  
(nonnegative-integer? 0)  ; Returns T  
(nonnegative-integer? 12.7) ; Returns NIL  
(nonnegative-integer? -12) ; Returns NIL
```

### 4.2.9 The positive-integer? Function

The `positive-integer?` function returns true if its argument is an integer strictly greater than 0.

#### Syntax

```
(positive-integer? x)
```

#### Examples

```
(positive-integer? 12) ; Returns T  
(positive-integer? 0)  ; Returns NIL  
(positive-integer? 12.7) ; Returns NIL  
(positive-integer? -12) ; Returns NIL
```

### 4.2.10 The product Function

The `product` function returns the product of the elements of a sequence, optionally transformed by a key function and restricted to a subsequence. An empty range yields 1.

#### Syntax

```
(product sequence &key key start end)
```

#### Arguments and Values

**sequence** A sequence of numbers.

**key** A function of one argument applied to each element before multiplication. The default is `identity`.

**start** The starting index (inclusive). The default is 0.

**end** The ending index (exclusive), or `nil` for the end of the sequence.

#### Examples

```
(product '(1 2 3 4 5)) ; Returns 120  
(product '(1 2 3 4 5) :start 2) ; Returns 60
```

### 4.2.11 The sawtooth-wave Function

The `sawtooth-wave` function is an alias for `fractional-value`.

### 4.2.12 The sum Function

The `sum` function returns the sum of the elements of a sequence, optionally transformed by a key function and restricted to a subsequence. An empty range yields 0.

#### Syntax

```
(sum sequence &key key start end)
```

#### Arguments and Values

**sequence** A sequence of numbers.

**key** A function of one argument applied to each element before addition. The default is `identity`.

**start** The starting index (inclusive). The default is 0.

**end** The ending index (exclusive), or `nil` for the end of the sequence.

### Examples

```
(sum (loop for i from 1 to 100 collect i)) ; Returns 5050
(sum '(1 2 3 4 5) :end 3)                 ; Returns 6
(sum '(1 2 3) :key (lambda (i) (* i 2))) ; Returns 12
```

#### 4.2.13 The unsigned-integer? Function

The `unsigned-integer?` function returns true if its argument is a nonnegative integer. It is an alias for `nonnegative-integer?`.

### Syntax

```
(unsigned-integer? x)
```

### Examples

```
(unsigned-integer? 12) ; Returns T
(unsigned-integer? 0) ; Returns T
(unsigned-integer? 12.7) ; Returns NIL
(unsigned-integer? -12) ; Returns NIL
```

## 4.3 Types

### 4.3.1 The nonnegative-float Type

The `nonnegative-float` type comprises floating-point numbers greater than or equal to 0.0.

### 4.3.2 The nonnegative-integer Type

The `nonnegative-integer` type comprises integers greater than or equal to 0.

### 4.3.3 The positive-float Type

The `positive-float` type comprises floating-point numbers strictly greater than 0.0.

#### **4.3.4 The positive-integer Type**

The `positive-integer` type comprises integers strictly greater than 0.



## Chapter 5

# The `sigma/os` Package

The `sigma/os` package provides a few helpers for talking to the operating system: reading files and running short scripts in other languages. The script runners use SBCL's `sb-ext:run-program` and are convenience wrappers, not full foreign interfaces.

### 5.1 Functions

#### 5.1.1 The `perl` Function

The `perl` function runs a short Perl snippet by invoking the interpreter at `*perl-path*` with `-e`. The arguments are concatenated into a single string of source before they are passed to Perl. Standard output is inherited from Lisp.

##### Syntax

```
(perl &rest code)
```

##### Examples

```
(perl "print_2+_2,_qq(\\n);")
```

#### 5.1.2 The `python` Function

The `python` function runs a short Python snippet by invoking the interpreter at `*python-path*` with `-c`. The arguments are concatenated into a single string of source before they are passed to Python. Standard output is inherited from Lisp.

**Syntax**

```
(python &rest code)
```

**Examples**

```
(python "print (2_+_2) ")
```

**5.1.3 The read-file Function**

The `read-file` function reads the entire contents of a file and returns them as a single string.

**Syntax**

```
(read-file filename)
```

**Arguments and Values**

*filename* A pathname designator for an existing file.

**Returns**

A string containing the full contents of the file.

**5.1.4 The read-lines Function**

The `read-lines` function reads the entire contents of a file and returns a list of its lines, without the terminating newlines.

**Syntax**

```
(read-lines filename)
```

**Arguments and Values**

*filename* A pathname designator for an existing file.

**Returns**

A list of strings, one per line.

### 5.1.5 The `ruby` Function

The `ruby` function runs a short Ruby snippet by invoking the interpreter at `*ruby-path*` with `-e`. The arguments are concatenated into a single string of source before they are passed to Ruby. Standard output is inherited from Lisp.

#### Syntax

```
(ruby &rest code)
```

#### Examples

```
(ruby "puts_2_+_2")
```

## 5.2 Parameters

### 5.2.1 The `*perl-path*` Parameter

The `*perl-path*` parameter holds the pathname of the Perl interpreter used by `perl`. The default is `/usr/bin/perl`.

### 5.2.2 The `*python-path*` Parameter

The `*python-path*` parameter holds the pathname of the Python interpreter used by `python`. The default is `/usr/bin/python`.

### 5.2.3 The `*ruby-path*` Parameter

The `*ruby-path*` parameter holds the pathname of the Ruby interpreter used by `ruby`. The default is `/usr/bin/ruby`.



## Chapter 6

# The sigma/probability Package

The `sigma/probability` package offers a small set of helpers for working with probabilities in the unit interval.

### 6.1 Macros

#### 6.1.1 The `decaying-probability?` Macro

The `decaying-probability?` macro performs a probabilistic test with `probability?`. When the test succeeds, it multiplies the place `probability` by `factor` (default 1/2) in place, so that successive calls become less likely, and returns `true`. When the test fails, it returns `nil` without changing `probability`.

#### Syntax

```
(decaying-probability? probability &optional factor)
```

#### Arguments and Values

***probability*** A place holding a value of type `probability`.

***factor*** The multiplicative decay applied after a successful trial. The default is 1/2.

#### Examples

```
(let ((p 1.0))
  (loop while (decaying-probability? p)
        collect p))
;; Collects a geometrically decaying sequence until a trial fails.
```

## 6.2 Functions

### 6.2.1 The probability? Function

The `probability?` function returns true with the given probability. It draws a uniform random number in  $[0,1)$  and returns true when that number is less than or equal to probability.

#### Syntax

```
(probability? probability)
```

#### Arguments and Values

**probability** A value of type probability.

#### Examples

```
(probability? 0.0) ; Always returns NIL
(probability? 1.0) ; Always returns T
(probability? 0.5) ; Returns T or NIL with equal chance
```

## 6.3 Types

### 6.3.1 The probability Type

The `probability` type comprises values suitable for use as probabilities: floating-point numbers in  $[0.0, 1.0]$ , the integers 0 and 1, and bits.

## Chapter 7

# The `sigma/random` Package

The `sigma/random` package provides helpers for random sampling, shuffling, and drawing from simple distributions.

### 7.1 Macros

#### 7.1.1 The `nshuffle` Macro

The `nshuffle` macro randomly shuffles its argument in place. It expands to an assignment of `(shuffle ...)` back into the place.

##### Syntax

```
(nshuffle argument)
```

##### Examples

```
(let ((xs '(1 2 3 4 5)))  
  (nshuffle xs)  
  xs) ; Returns the same elements in a random order
```

### 7.2 Functions

#### 7.2.1 The `gauss` Function

The `gauss` function draws a sample from a Gaussian (normal) distribution with mean `mu` and standard deviation `sigma`. The implementation

uses the Box-Muller method, caching the second sample of each pair between calls.

**Syntax**

```
(gauss mu sigma)
```

**Arguments and Values**

*mu* The mean of the distribution.

*sigma* The standard deviation of the distribution.

**7.2.2 The random-argument Function**

The `random-argument` function returns one of its arguments chosen uniformly at random, or `nil` if no arguments are given.

**Syntax**

```
(random-argument &rest rest)
```

**Examples**

```
(random-argument 'rock 'paper 'scissors)
```

**7.2.3 The coin-toss Function**

The `coin-toss` function returns `t` or `nil` with equal probability, like a fair coin toss.

**Syntax**

```
(coin-toss)
```

**7.2.4 The random-in-range Function**

The `random-in-range` function returns a random number in the half-open interval  $[lower, upper)$ . Either bound may be a sequence, in which case the most extreme member of that sequence is used (`maximum` for the lower bound, `minimum` for the upper). If the lower bound is greater than the upper, the bounds are swapped. If they are equal, that common value is returned.



**Syntax**

```
(random-in-range lower upper)
```

**Examples**

```
(random-in-range 0 10)      ; A number in [0, 10)
(random-in-range 0.0 1.0)    ; A float in [0.0, 1.0)
```

**7.2.5 The random-in-ranges Function**

The `random-in-ranges` function, given one or more restricting ranges each as a two-element list, returns a random number in a range common to all of them.

**Syntax**

```
(random-in-ranges &rest ranges)
```

**Examples**

```
(random-in-ranges '(0 10) '(5 15)) ; A number in a common subrange
```

**7.2.6 The random-range Function**

The `random-range` function returns a two-element list (*lo hi*) of random bounds within [*lower*, *upper*). Without `:containing`, the two bounds are independent samples ordered so that  $lo \leq hi$ . With `:containing` (a number or sequence), the result is chosen so that the returned range covers that value or those values.

**Syntax**

```
(random-range lower upper &key containing)
```

**7.2.7 The randomize-array Function**

The `randomize-array` function fills an array in place with values from (`random argument-for-random`) and returns the array.

**Syntax**

```
(randomize-array array argument-for-random)
```

### 7.2.8 The random-array Function

The `random-array` function returns a new array of the given dimensions whose contents are randomized as by `randomize-array`.

#### Syntax

```
(random-array dimensions argument-for-random)
```

#### Examples

```
(random-array '(3 3) 10) ; A 3-by-3 array of integers in [0, 10)
```

## 7.3 Generics

### 7.3.1 The random-element Generic

The `random-element` generic returns a randomly chosen element from a sequence, or `nil` if the sequence is empty. Methods are provided for lists and arrays.

#### Syntax

```
(random-element sequence)
```

#### Examples

```
(random-element '(a b c d))  
(random-element #(1 2 3 4 5))
```

### 7.3.2 The shuffle Generic

The `shuffle` generic returns a new container with the same elements as its argument in random order. The original is not modified; use `nshuffle` to shuffle a place in place. Methods are provided for lists and arrays.

#### Syntax

```
(shuffle container)
```

**Examples**

```
(shuffle '(1 2 3 4 5))  
(shuffle #(a b c d e))
```



## Chapter 8

# The `sigma/sequence` Package

The `sigma/sequence` package provides utilities for lists, vectors, and other sequences: extrema, sorting by a parallel key, slicing, splitting, and a few destructive helpers.

### 8.1 Macros

#### 8.1.1 The `nconcf` Macro

The `nconcf` macro destructively concatenates `list-2` onto `list-1` and stores the result back into the place `list-1`, analogous to `nconc` with assignment.

##### Syntax

```
(nconcf list-1 list-2)
```

##### Examples

```
(let ((a '(1 2 3))
      (b '(4 5 6)))
  (nconcf a b)
  a) ; Returns '(1 2 3 4 5 6)
```

#### 8.1.2 The `set-nthcdr` Macro

The `set-nthcdr` macro sets the *n*th `cdr` of a list to a new value. When *n* is 0, it sets the list place itself. On implementations that allow it, this is also installed as the `setf` expander for `nthcdr`.

**Syntax**

```
(set-nthcdr n list new-value)
```

**8.2 Functions****8.2.1 The arefable? Function**

The `arefable?` function returns true if `position` is a valid multidimensional index list for `array` via `aref`: the same rank as the array, with each coordinate in range for its dimension.

**Syntax**

```
(arefable? array position)
```

**8.2.2 The array-values Function**

The `array-values` function returns a list of the values in an array at the specified positions. Each position is a list of subscripts suitable for `aref`.

**Syntax**

```
(array-values array positions)
```

**Examples**

```
(array-values #2A((1 2) (3 4))
              '((0 0) (1 1))) ; Returns '(1 4)
```

**8.2.3 The nth-from-end Function**

The `nth-from-end` function is similar to `nth`, but counts from the end of the list. Index 0 is the last element.

**Syntax**

```
(nth-from-end n list)
```

**Examples**

```
(nth-from-end 0 '(0 1 2 3 4 5 6 7 8 9 10)) ; Returns 10
(nth-from-end 3 '(0 1 2 3 4 5 6 7 8 9 10)) ; Returns 7
```

### 8.2.4 The nthable? Function

The `nthable?` function returns true if  $n$  is a valid index for `list` via `nth`: a nonnegative integer strictly less than the length of the list.

#### Syntax

```
(nthable? n list)
```

### 8.2.5 The sequence? Function

The `sequence?` function returns true if its argument is of type `sequence` (a list or vector).

#### Syntax

```
(sequence? sequence)
```

### 8.2.6 The empty-sequence? Function

The `empty-sequence?` function returns true if its argument is a sequence that contains no elements. For arrays, any dimension of size zero counts as empty.

#### Syntax

```
(empty-sequence? sequence)
```

#### Examples

```
(empty-sequence? '()) ; Returns T  
(empty-sequence? #()) ; Returns T  
(empty-sequence? '(a)) ; Returns NIL
```

### 8.2.7 The join-symbol-to-all-following Function

This function takes a symbol and a list, and for every occurrence of the symbol in the list, it joins it to the item following it.

#### Syntax

```
(join-symbol-to-all-following symbol list)
```

**Examples**

```
(join-symbol-to-all-following :# '(:# 10 :# 20 :# 30))
;; Returns '(:#10 :#20 :#30)
```

**Affected By**

\*print-escape\*, \*print-radix\*, \*print-base\*, \*print-circle\*,  
 \*print-pretty\*, \*print-level\*, \*print-length\*, \*print-case\*,  
 \*print-gensym\*, \*print-array\*.

**8.2.8 The join-symbol-to-all-preceding Function**

This function takes a symbol and a list, and for every occurrence of the symbol in the list, it joins it to the item preceding it.

**Syntax**

```
(join-symbol-to-all-preceding symbol list)
```

**Examples**

```
(join-symbol-to-all-preceding :% '(10 :% 20 :% 30 :%))
;; Returns '(:10% :20% :30%)
```

**Affected By**

\*print-escape\*, \*print-radix\*, \*print-base\*, \*print-circle\*,  
 \*print-pretty\*, \*print-level\*, \*print-length\*, \*print-case\*,  
 \*print-gensym\*, \*print-array\*.

**8.2.9 The list-to-vector Function**

The `list-to-vector` function takes a list and returns an equivalent vector.

**Syntax**

```
(list-to-vector list)
```

**Examples**

```
(list-to-vector '(1 2 3)) ; Returns #(1 2 3)
```



### 8.2.10 The `max*` Function

The `max*` function is a shortcut for `max`. It takes in one or more lists and finds the maximum value within all of them, so that you need not manually use `apply` and `concatenate`.

#### Syntax

```
(max* &rest lists)
```

#### Examples

```
(max* '(1 2 3 100 4 5)) ; Returns 100
(max* '(1 2 3 4)
      '(5 6 99 7)
      '(8 9 10)) ; Returns 99
```

### 8.2.11 The `min*` Function

The `min*` function is a shortcut for `min`. It takes in one or more lists and finds the minimum value within all of them, so that you need not manually use `apply` and `concatenate`.

#### Syntax

```
(min* &rest lists)
```

#### Examples

```
(min* '(1 2 3 -100 4 5)) ; Returns -100
(min* '(1 2 3 4)
      '(5 6 -99 7)
      '(8 9 10)) ; Returns -99
```

### 8.2.12 The `set-equal` Function

The `set-equal` function returns true if two lists contain the same elements when viewed as sets, using an optional `key` and either `test` or `test-not` for element comparison. Order and duplicate multiplicity are ignored.

#### Syntax

```
(set-equal list-1 list-2 &key key test test-not)
```

**Examples**

```
(set-equal '(1 2 3) '(3 2 1))      ; Returns T
(set-equal '(1 2) '(1 2 3))      ; Returns NIL
(set-equal '("a" "b") '("A" "B"))
      :test #'string-equal)      ; Returns T
```

**8.2.13 The simple-vector-to-list Function**

The `simple-vector-to-list` function takes a vector and returns an equivalent list, reading elements with `svref`.

**Syntax**

```
(simple-vector-to-list vector)
```

**8.2.14 The sort-order Function**

The `sort-order` function returns the indices that would sort the sequence. Sorting those indices by the corresponding elements under `predicate` and `key` yields the sorted order of the original sequence.

**Syntax**

```
(sort-order sequence predicate &key key)
```

**Examples**

```
(sort-order '(30 10 20) #'<) ; Returns '(1 2 0)
```

**8.2.15 The the-last Function**

The `the-last` function returns the last element of a list (the `car` of its last `cons`), or `nil` if the list is empty.

**Syntax**

```
(the-last list)
```

**Examples**

```
(the-last '(a b c d)) ; Returns D
(the-last '())         ; Returns NIL
```

### 8.2.16 The vector-to-list Function

The `vector-to-list` function takes a vector and returns an equivalent list.

#### Syntax

```
(vector-to-list vector)
```

#### Examples

```
(vector-to-list #(1 2 3)) ; Returns '(1 2 3)
```

## 8.3 Generics

### 8.3.1 The best Generic

The `best` generic returns the “best” element of a sequence according to a predicate and optional key. This is equivalent to taking the first element after sorting the sequence with the same predicate and key, but runs in linear time. Methods are provided for lists and vectors. An empty sequence yields `nil`.

#### Syntax

```
(best sequence predicate &key key)
```

#### Examples

```
(best '(3 1 4 1 5) #'<) ; Returns 1  
(best '(3 1 4 1 5) #'>) ; Returns 5
```

### 8.3.2 The minimum Generic

The `minimum` generic returns the minimum element of a sequence, optionally after applying a key, considering only the subsequence from `start` to `end`.

#### Syntax

```
(minimum sequence &key key start end)
```

**Examples**

```
(minimum '(3 1 4 1 5)) ; Returns 1
```

**8.3.3 The minimum? Generic**

The `minimum?` generic returns true if the element at position in the sequence is a minimum under key within the subsequence from start to end. When position is omitted, it defaults to the last index.

**Syntax**

```
(minimum? sequence &key position key start end)
```

**8.3.4 The maximum Generic**

The `maximum` generic returns the maximum element of a sequence, optionally after applying a key, considering only the subsequence from start to end.

**Syntax**

```
(maximum sequence &key key start end)
```

**Examples**

```
(maximum '(3 1 4 1 5)) ; Returns 5
```

**8.3.5 The maximum? Generic**

The `maximum?` generic returns true if the element at position in the sequence is a maximum under key within the subsequence from start to end. When position is omitted, it defaults to the last index.

**Syntax**

```
(maximum? sequence &key position key start end)
```

**8.3.6 The sort-on Generic**

The `sort-on` generic sorts one sequence according to a parallel ordering sequence. Corresponding elements are paired; the pairs are sorted by the predicate on the ordering values (optionally through a key), and the sorted elements of the sequence to sort are returned. Methods accept lists and vectors in either role.

**Syntax**

```
(sort-on sequence-to-sort ordering-sequence predicate &key
key)
```

**Examples**

```
(sort-on '(a b c) '(3 1 2) #'<) ; Returns '(b c a)
```

**8.3.7 The slice Generic**

The `slice` generic returns a modular subset of a sequence, taking every `slice`-th element (`slice` defaults to 1). The stride may be any positive rational number. Methods are provided for lists and vectors.

**Syntax**

```
(slice sequence &optional slice)
```

**Examples**

```
(slice '(1 2 3 4 5 6 7 8 9) 2) ; Returns '(1 3 5 7 9)
(slice #(1 2 3 4 5 6 7 8 9) 2) ; Returns #(1 3 5 7 9)
```

**8.3.8 The split Generic**

The `split` generic splits a sequence into subsequences wherever an element is a member of separators (under `key` and `test`). When `remove-separators?` is true (the default), separator elements are omitted from the results. Methods are provided for lists (and, in the string package, for strings).

**Syntax**

```
(split sequence separators &key key test remove-separators?)
```

**Examples**

```
(split '(1 2 0 3 4 0 5) '(0))
;; Returns '((1 2) (3 4) (5))
```

### 8.3.9 The worst Generic

The `worst` generic returns the “worst” element of a sequence according to a predicate and optional key. This is equivalent to taking the last element after sorting the sequence with the same predicate and key, but runs in linear time. Methods are provided for lists and vectors. An empty sequence yields `nil`.

#### Syntax

```
(worst sequence predicate &key key)
```

#### Examples

```
(worst '(3 1 4 1 5) #'<) ; Returns 5  
(worst '(3 1 4 1 5) #'>) ; Returns 1
```

## Chapter 9

# The `sigma/string` Package

The `sigma/string` package contains useful tools for working with strings: ranges of characters, joining and repeating, trimming whitespace, and converting other objects to text.

### 9.1 Constants

#### 9.1.1 The `+whitespace+` Parameter

The `+whitespace+` constant is a list of characters treated as whitespace by the trimming functions: `#\Space`, `#\Newline`, `#\Backspace`, `#\Tab`, `#\Linefeed`, `#\Page`, `#\Return`, and `#\Rubout`.

### 9.2 Functions

#### 9.2.1 The `character-range` Function

The `character-range` function returns a list of characters from the *start* to the *end* character. Note that this returns a list, not a string. The endpoints may be given in either order; the result is always ascending.

##### Syntax

```
(character-range start end)  $\Rightarrow$  ' (start ... end)
```

**Arguments and Values**

**start** The character to start the range with, inclusive.

**end** The character to end the range with, inclusive.

**Examples**

```
(character-range #\a #\e) => ' (#\a #\b #\c #\d #\e)
(character-range #\e #\a) => ' (#\a #\b #\c #\d #\e)
```

**9.2.2 The character-ranges Function**

The `character-ranges` function is a convenience wrapper for `character-range`, concatenating several calls and making the resultant list contain only unique instances, sorted in ascending order.

**Syntax**

```
(character-ranges start1 end1 ... => ' (character1 ...)
```

**Arguments and Values**

**start<sub>n</sub>** The character to start the *n*th range with, inclusive.

**end<sub>n</sub>** The character to end the *n*th range with, inclusive.

**Examples**

```
(character-ranges #\a #\c #\x #\z) => ' (#\a #\b #\c #\x #\y
#\z)
(character-ranges #\a #\c #\a #\c) => ' (#\a #\b #\c)
```

**9.2.3 The escape-tildes Function**

The `escape-tildes` function returns a copy of a string in which every tilde (`#\~`) is doubled, suitable for use as a `format` control string that should print tildes literally.

**Syntax**

```
(escape-tildes string)
```

**Examples**

```
(escape-tildes "foo_bar") ; Returns "foo bar"
(escape-tildes "foo~bar") ; Returns "foo~~bar"
```



### 9.2.4 The replace-char Function

The `replace-char` function replaces every instance of `from-char` in `string` with `to-char`. The string is modified in place and returned.

#### Syntax

```
(replace-char string from-char to-char)
```

#### Examples

```
(replace-char (copy-seq "banana") #\textbackslash a #\textbackslash o)
;; Returns "bonono"
```

### 9.2.5 The strcat Function

The `strcat` function takes any number of arguments, converts each to a string with `to-string`, and concatenates the results. The symbol `string-concatenate` is an alias for this function.

#### Syntax

```
(strcat &rest rest)
(string-concatenate &rest rest)
```

#### Examples

```
(strcat "foo" "bar")           ; Returns "foobar"
(strcat "foo" 123 "bar")       ; Returns "foo123bar"
(strcat)                       ; Returns ""
(strcat 1 2 3 4)               ; Returns "1234"
```

### 9.2.6 The string-concatenate Function

The `string-concatenate` function is an alias for `strcat`.

### 9.2.7 The strmult Function

The `strmult` function concatenates its string arguments (via `strcat`) and repeats that result `count` times, returning one combined string. When `count` is less than 1, it returns the empty string.

#### Syntax

```
(strmult count &rest strings)
```

**Examples**

```
(strmult 3 "ab")      ; Returns "ababab"  
(strmult 2 "x" "y")  ; Returns "xyxy"  
(strmult 0 "nope")   ; Returns ""
```

**9.2.8 The string-join Function**

The `string-join` function joins a list of strings into one string, inserting `connecting-string` between each adjacent pair. A single string may be passed and is returned unchanged aside from the single-element join path. The connecting string defaults to the empty string.

**Syntax**

```
(string-join strings &optional connecting-string)
```

**Examples**

```
(string-join '("a" "b" "c") "-") ; Returns "a-b-c"  
(string-join '("a" "b" "c"))      ; Returns "abc"
```

**9.2.9 The stringify Function**

The `stringify` function takes an argument of any type and converts it to a string. This produces the string as from the `~A` directive to `format`. Also see `to-string`.

**Syntax**

```
(stringify argument)
```

**Examples**

```
(stringify 12) ; Returns "12"
```

**9.2.10 The string-trim-whitespace Function**

The `string-trim-whitespace` function removes whitespace from both ends of a string, using the characters in `+whitespace+`.

**Syntax**

```
(string-trim-whitespace string)
```

**Examples**

```
(string-trim-whitespace "___foo___") ; Returns "foo"
```

**9.2.11 The string-left-trim-whitespace Function**

The `string-left-trim-whitespace` function removes whitespace from the left side only of a string.

**Syntax**

```
(string-left-trim-whitespace string)
```

**Examples**

```
(string-left-trim-whitespace "___foo___") ; Returns "foo  "
```

**9.2.12 The string-right-trim-whitespace Function**

The `string-right-trim-whitespace` function removes whitespace from the right side only of a string.

**Syntax**

```
(string-right-trim-whitespace string)
```

**Examples**

```
(string-right-trim-whitespace "___foo___") ; Returns "  foo"
```

**9.2.13 The to-string Function**

The `to-string` function converts common types of things into a string. It handles some special cases more usefully than `stringify` for most user-facing output: `nil` becomes the empty string, and symbols become their downcased names.

**Syntax**

```
(to-string s)
```

**Examples**

```
(to-string nil)      ; Returns ""  
(to-string :foo)     ; Returns "foo"  
(to-string "hello"); Returns "hello"
```

## 9.3 Methods

### 9.3.1 The split Methods

The `split` method on strings splits a string into a list of substrings on separators. Separators may be a single separator or a sequence of them. The `key` and `test` arguments control membership tests, and `remove-separators?` controls whether separator text is kept. The implementation coerces the string to a list, delegates to the list method of `split` from `sigma/sequence`, and coerces each piece back to a string.

**Syntax**

```
(split string separators &key key test remove-separators?)
```

**Examples**

```
(split "a,b,c" #\, ) ; Returns '("a" "b" "c")
```

## Chapter 10

# The `sigma/time-series` Package

The `sigma/time-series` package supports simple time series and multivariate time series (time multiseries), together with geometric helpers such as raster lines through grids, norms, and distances.

A *time series* is a nonempty list whose elements share a common type. A *time multiseries* is a nonempty list of arrays of equal dimensions, where each array holds the data for one time step.

### 10.1 Macros

#### 10.1.1 The `snap-index` Macro

The `snap-index` macro wraps an index into the half-open interval  $[0, bound)$  by adding or subtracting `bound` as needed. It modifies the place `index` in place.

##### Syntax

```
(snap-index index bound)
```

### 10.2 Functions

#### 10.2.1 The `array-raster-line` Function

The `array-raster-line` function returns a one-dimensional list of values from an array along the straight-line path from `start-point` to `end-point`. The array may be of any rank. Points are generated by `raster-line` and read with `array-values`.

**Syntax**

```
(array-raster-line array start-point end-point &key coordinate-assertion
from-start from-end)
```

**10.2.2 The distance Function**

The `distance` function calculates the distance between two points as the norm of their coordinate-wise difference. The optional power is as for `norm`.

**Syntax**

```
(distance initial-point final-point &optional power)
```

**Examples**

```
(distance '(0 0) '(3 4)) ; Returns 5.0
```

**10.2.3 The norm Function**

The `norm` function returns the mathematical vector norm of a sequence. For the infinity norm, pass `:infinity` as the power.

**Syntax**

```
(norm sequence &optional power)
```

**Arguments and Values**

**sequence** A sequence of numbers.

**power** A positive number, or `:infinity`. The default is 2 (the Euclidean norm).

**Examples**

```
(norm '(3 4)) ; Returns 5.0
(norm '(3 4) :infinity) ; Returns 4
```

### 10.2.4 The raster-line Function

The `raster-line` function returns the discrete points on a straight line from `start-point` to `end-point` in an  $n$ -dimensional integer grid. It is derived from the three-dimensional scan-conversion algorithm of Kaufman and Shimony, generalized to any nonnegative dimension. Optional `from-start` and `from-end` trim points from either end of the basic path.

#### Syntax

```
(raster-line start-point end-point &key coordinate-assertion
  from-start from-end)
```

#### Arguments and Values

**`start-point`** A list of coordinates.

**`end-point`** A list of coordinates of the same length.

**`coordinate-assertion`** A predicate applied to each coordinate (default `integerp`).

**`from-start`** Nonnegative integer offset from the start.

**`from-end`** Integer offset from the end.

### 10.2.5 The similar-points? Function

The `similar-points?` predicate returns true if two points are lists of equal length whose coordinates all satisfy an optional assertion (default `numberp`).

#### Syntax

```
(similar-points? p q &optional coordinate-assertion)
```

### 10.2.6 The time-series? Function

The `time-series?` predicate returns true if its argument could be a time series: a nonempty list whose every element is of the given element type (default `t`).

#### Syntax

```
(time-series? time-series &optional element-type)
```

**Examples**

```
(time-series? '(1.0 2.0 3.0) 'float) ; Returns T
(time-series? '())                  ; Returns NIL
```

**10.2.7 The time-multiseries? Function**

The `time-multiseries?` predicate returns true if its argument is a time multiseries: a nonempty list of arrays of equal dimensions.

**Syntax**

```
(time-multiseries? time-multiseries)
```

**10.2.8 The tmsref Function**

The `tmsref` function works like `aref`, but for a time series or multiseries. For a one-dimensional time series (a list), it returns the element at the given time. For a multiseries, it returns the array element at position within the array for that time step.

**Syntax**

```
(tmsref time-multiseries time &rest position)
```

**Examples**

```
(tmsref '(10 20 30) 1) ; Returns 20
```

**10.2.9 The tms-dimensions Function**

The `tms-dimensions` function works like `array-dimensions`, but for a time series or multiseries. The first dimension listed is the time dimension.

**Syntax**

```
(tms-dimensions time-multiseries)
```

**10.2.10 The tms-values Function**

The `tms-values` function returns a list of the values in a time series or multiseries at the specified positions. Each position is a list whose first element is the time index; further elements are spatial subscripts for a multiseries. A bare integer is treated as a one-element position list.



**Syntax**

(tms-values *time-multiseries* *positions*)

**10.3 Types****10.3.1 The time-multiseries Type**

The `time-multiseries` type comprises objects that satisfy `time-multiseries?`.



# Chapter 11

## The `sigma/truth` Package

The `sigma/truth` package provides a few small tools for converting and negating truth values.

### 11.1 Functions

#### 11.1.1 The `[?]` Function

The `[?]` function is Knuth's truth function. It converts its argument to 1 for true and 0 for false. In other words, `nil` maps to 0 and everything else maps to 1.

##### Syntax

```
([?] x)
```

##### Examples

```
([?] nil) ; Returns 0  
([?] t)   ; Returns 1  
([?] 12)  ; Returns 1
```

#### 11.1.2 The `toggle` Function

The `toggle` function returns the boolean negation of its argument: `nil` if the argument is true, and `t` if the argument is false.

##### Syntax

```
(toggle x)
```

**Examples**

```
(toggle nil) ; Returns T
(toggle t)   ; Returns NIL
(toggle 12)  ; Returns NIL
```

## 11.2 Generics

### 11.2.1 The ? Generic

The `?` generic converts a generalized boolean into a strict boolean: `nil` stays `nil`, and any non-`nil` value becomes `t`.

**Syntax**

```
(? x)
```

**Examples**

```
(? nil) ; Returns NIL
(? t)   ; Returns T
(? 12)  ; Returns T
(? '()) ; Returns NIL
```

## Chapter 12

# The sigma Package

The `sigma` package is the umbrella package for the library. It uses every subordinate package and re-exports a short list of entry points for loading the whole system into the current package.

### 12.1 Variables

#### 12.1.1 The `*sigma-packages*` Variable

The `*sigma-packages*` variable holds a list of all Sigma package designators. `use-all-sigma` walks this list. The default value is

```
(:sigma/behave
 :sigma/control
 :sigma/hash
 :sigma/numeric
 :sigma/os
 :sigma/probability
 :sigma/random
 :sigma/sequence
 :sigma/string
 :sigma/time-series
 :sigma/truth
 :sigma)
```

### 12.2 Functions

#### 12.2.1 The `use-all-sigma` Function

The `use-all-sigma` function calls `use-package` on every package in `*sigma-packages*` from the current package, making all of their external symbols available without package prefixes. This will pollute

`common-lisp-user` if that is your current package; that is intentional for interactive use.

**Syntax**

```
(use-all-sigma)
```

**Examples**

```
(asdf:load-system :sigma)
(sigma:use-all-sigma)
(sum (loop for i from 1 to 100 collect i)) ; Returns 5050
```

# Bibliography

- [1] GRAHAM, P. *On Lisp*. Prentice-Hall, 1993.
- [2] GRAHAM, P. *ANSI Common Lisp*. Prentice-Hall, 1995.